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The models were evaluated using classification accuracy and categorical cross-entropy loss. Accuracy was chosen as the primary metric because the task is a single label multi-class classification problem, where each input belongs to exactly one of six classes. It provides a clear and interpretable measure of overall model performance and enables straightforward comparison across architectures. Categorical cross-entropy loss was also used to assess how well the models estimate class probabilities. Unlike accuracy, which only considers correct predictions, loss reflects the confidence of predictions and is useful for monitoring training behaviour.

In addition, confusion matrices were used to analyse class-level performance. These provide insight into which classes are frequently misclassified and help identify systematic errors, particularly between semantically similar categories such as DESC and ENTY.

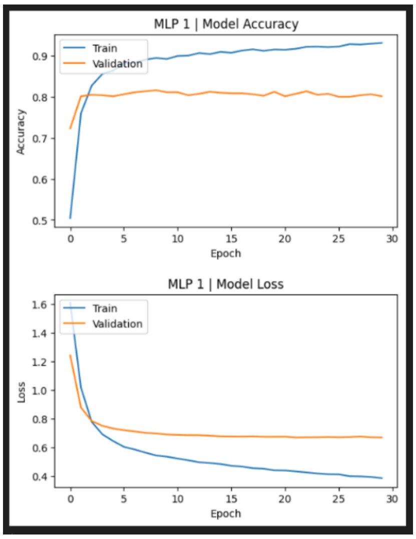
Several techniques and hyperparameter settings were explored to improve model performance. First, two different text representations were investigated: TF-IDF features and pre-trained word embeddings (GloVe).

For the neural architectures, various hyperparameters were adjusted, including network size dropout rates, regularisation strength, sequence length, and vocabulary size. Some parameters were kept the same to ensure fair results. For instance, the epoch value for MLP1 and MLP2 is 30 and the value is 10 for CNN and RNN. A 0.001 regularizer was used for all models.

**Result summary:**

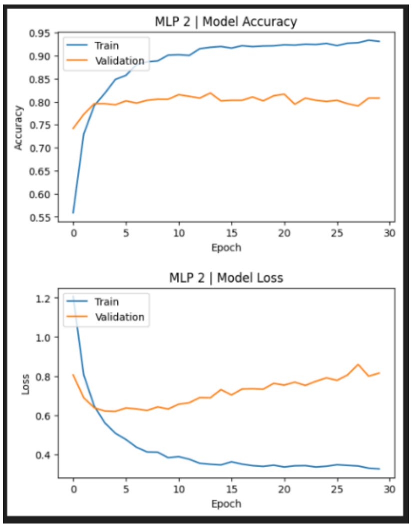
| Model | Test Accuracy | Test Loss |
|---|---|---|
| MLP1 | 0.8380 | 0.6391 |
| MLP2 | 0.8760 | 0.5516 |
| CNN | 0.9020 | 0.3354 |
| RNN | 0.8640 | 0.4468 |

The above results show that the CNN model achieved the highest test accuracy and lowest test loss, marking it as the strongest model in this assignment.




TF-IDF is a strong baseline for short-text classification because it captures which words are important in a question. For TREC, question words such as who, where, when, and how many are often predictive of the label so an MLP over TF-IDF features is expected to perform reasonably well even without sequence modelling.

This model achieved 83.80% test accuracy which is a solid baseline, however, this approach ignores word order and

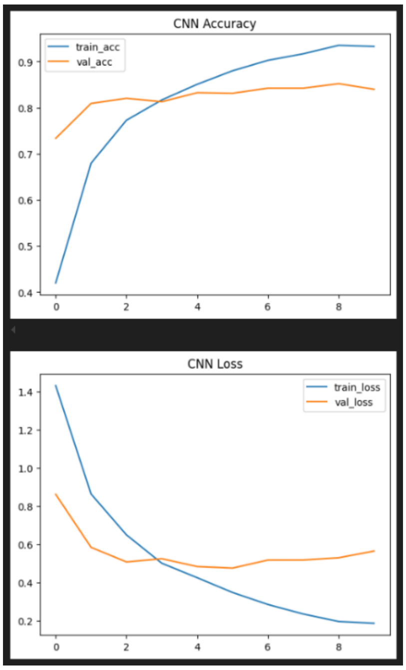



deeper semantic relationships between words. As a result, it is less effective than the embedding-based approaches.

The training history suggests that the model learned quickly, but performance plateaued earlier than the more advanced models. This indicates that TF-IDF provides useful, but limited information, for this task.

Unlike TF-IDF, pre-trained embeddings encode semantic similarity between words. This means that semantically related words can have similar vector representations, which should help the model generalise better.

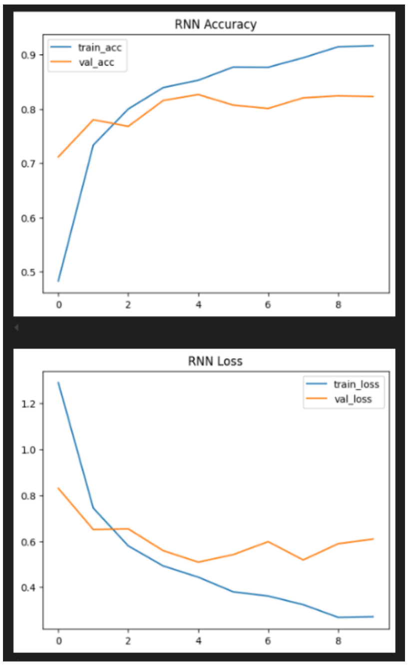
The use of pre-trained embeddings was expected to improve performance because the TREC dataset is relatively small. This expectation proved true, with this model achieving a 87.60% test accuracy, clearly outperforming the TF-IDF MLP, indicating that semantic word representations were beneficial for this assignment.




CNNs are effective for text classification because they can learn local patterns such as key phrase. The CNN produced the best overall performance, reaching 90.20% test accuracy with the lowest test loss, 0.3354.

Compared with the MLP models, the CNN benefits from both semantic embeddings and sequence aware feature extraction. Compared with the RNN, the CNN appears to capture the most useful information more efficiently, without having to model longer sequential dependencies that may not be necessary for such short inputs.

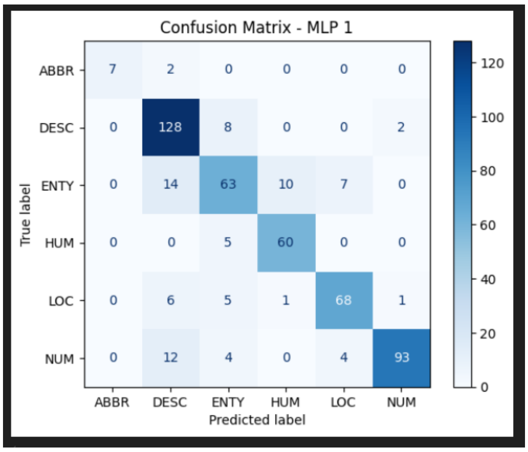
The training history also suggests some overfitting in later epochs, since training accuracy continued to rise while validation performance became less stable but even so, the final test accuracy remained the highest among all models.

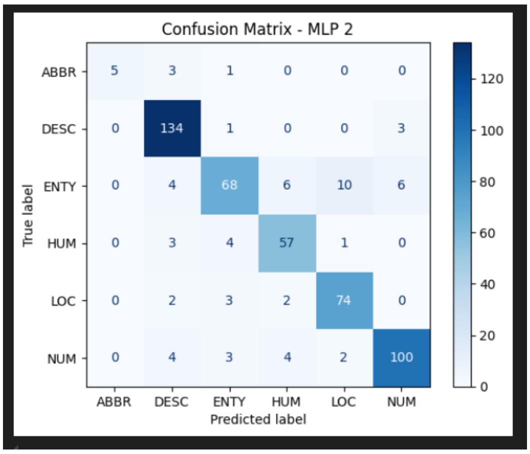


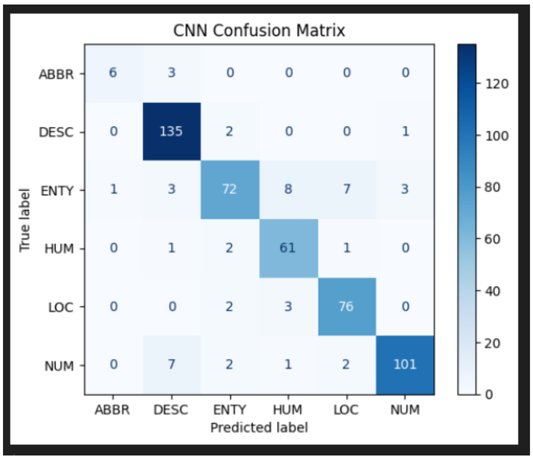

RNNs, especially LSTMs, are designed to model sequential dependencies which is useful when the meaning of a sentence depends on word order. For text classification in general, LSTMs are expected to perform well because they can capture patterns that simpler bag-of-words models may miss.

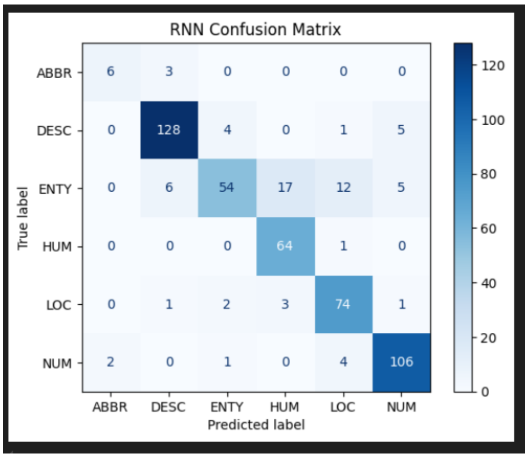
The RNN achieved 86.40% test accuracy, which is better than MLP 1 but lower than both MLP 2 and the CNN. This suggests that while sequential modelling is helpful, it was not the most effective approach for this specific dataset. One likely reason is that TREC questions are usually short and formulaic, meaning that long-distance dependencies are less important than recognising a few key local patterns.

The validation results also suggest some instability and possible overfitting across epochs, with validation accuracy fluctuating more than in the CNN model.




The confusion matrices reveal consistent patterns across all models. Classes such as HUM, LOC, and NUM are classified with high accuracy due to strong lexical cues (e.g. “who”, “where”, “how many”). The main source of error is confusion between DESC and ENTY, which share similar structures (e.g. “What is…”). The TF-IDF MLP shows the highest confusion between these classes, reflecting its inability to capture semantic nuance.

The embedding-based MLP reduces this issue, while the CNN shows the clearest class separation. The RNN performs moderately well but does not improve upon the CNN, likely due to the short length of the input sequences. Overall, the confusion matrices highlight that performance differences are driven by how well models handle semantic similarity and local context.

Overall, the CNN model is the best-performing model due to its superior accuracy and ability to capture key phrase-level features. It provides the best balance between performance and complexity, making it the most suitable choice for deployment in a question classification system.

The project successfully implemented and compared multiple neural architectures, achieving strong performance across all models. However, several limitations remain. For one, the dataset is relatively small, increasing the risk of overfitting. This problem of overfitting could be managed better if early stopping was implemented to prevent overfitting, but I decided it would be best not to implement this in order to keep the results fair for the purpose of this assignment.

Another limitation is that the evaluation relies primarly on accuracy, but additional metrics such as F1 could provide further insights into model performance and evaluation.

Another strategy that I would have liked to implement would have been a more advanced pre-trained mode such as BERT, which could significantly improve performance by providing context aware embeddings that capture deeper linguistic relationships.

Lastly, hyperparameter tuning was limited and could be further explored. Parameters such as learning rate, batch size, kernel sizes, dropout rates, and sequence length could be systematically optimised using techniques such as grid search to further improve model performance.

While the CNN model achieves a high accuracy rate, given that a small data set was used and that the evaluation relies primarly on accuracy, I would be of the opinion that this model is not suffciently reliable for fully automated deployment. For these reasons, I would propose that this model be used as a decision support tool within a human-in-the-loop system where human expertise is integrated into the training, validation, or inference phases of a CNN model.

In this approach, the model generates a predicted class along with a confidence score for each input question. Predictions with high confidence can be accepted automatically, while those with low or uncertain confidence are flagged for human review. This reduces the workload for human moderators while maintaining quality control.

Additionally, human feedback on incorrect or uncertain predictions can be collected and used to continuously improve the model through retraining, creating an iterative system where the model assists in classification while gradually becoming more accurate over time, potentionally allowing the model to surpass the human-in-the-loop system, allowing it be used without human supervision.